SAME PLACE, FROM SPACE 2016

ESMT · DATA FOR GOOD · FRANKFURT ZOOLOGICAL SOCIETY

What should FZS track every year?

Five numbers to see illegal gold mining in the Amazon earlier, starting with Peru, and to act where it matters.

Group 5 · Giant River Turtle · October 2026

Likitha Yenuga · Laura Pérez · Pietro Maran · Erion Vathi · Santiago Sierra

Background: one place near the Madre de Dios river, outside protected areas, seen by satellite each year from 2016 to 2025. Sentinel 2 cloudless by EOX IT Services (contains modified Copernicus Sentinel data).

BEFORE WE START · OUR ANIMAL

Two facts about the giant river turtle

73 kg

The biggest river turtle in South America. The record female weighed 73 kg, about as much as an adult person.

11

Babies call from inside the egg. Mothers answer and swim away with them. Scientists recorded 11 types of sounds.

> They lay their eggs on **river sand beaches**. Gold miners dig the same beaches and rivers.

FAO species profile, Podocnemis expansa; Ferrara, Vogt and Sousa Lima (2013), Journal of Comparative Psychology

WHY TIMING MATTERS

The yearly satellite maps show new mining many months later

2021 to 2023

the surge: forest lost to mining more than doubles

March 2023

the first MapBiomas Peru map, with data up to 2021

5 to 17

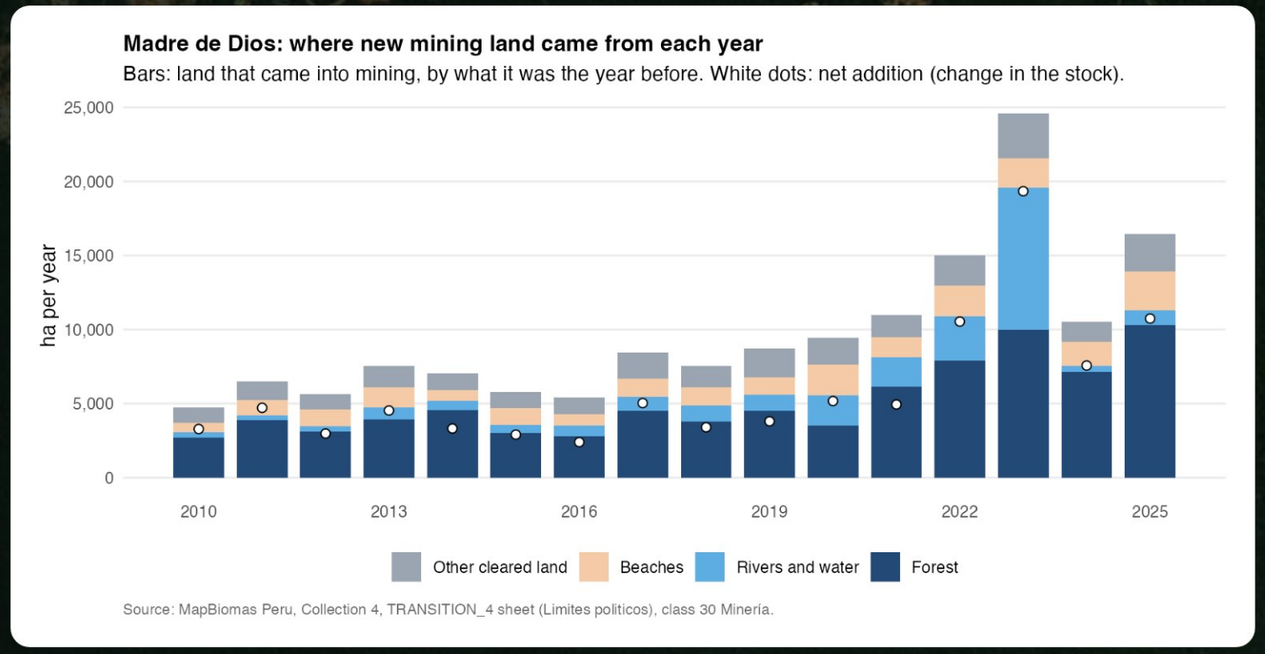
months late, depending on the year

> So we looked for numbers that **move earlier** and **say where**, to add to the maps and to field work.

Release dates: Panamericana (Collection 1), MapBiomas Peru (Collections 2 and 3), Caretas (Collection 4). Forest lost: MapBiomas Peru C4.

WHAT THE DATA SHOWED FIRST · MADRE DE DIOS

The satellite maps confuse old mining ponds with water



Mining land called **water** in 2021 and 2022 — **4,595 ha**

Then called mining again in 2023: 39% of all new mining land — **9,616 ha**

The same jump in the rest of Peru in 2023, so the map itself likely changed — **× 2.7**

Decision: we only count forest that became mining. Trees turning into bare ground are easy to see from space.

MapBiomas Peru Collection 4, transitions. Ponds are hard to classify: Caballero Espejo et al. 2018; Camalan et al. 2022.

HOW WE CHOSE THE KPIS

Three speeds, five tests

EARLY · MONTHLY

Before new mining appears

Gold price and gold trade. Pressure shows up months before the maps.

KPI 1 · gold price
KPI 2 · gold gap

NOW · QUARTERLY

Where mining is happening

Satellite detections, available days after each quarter ends.

KPI 3 · FZS landscapes
KPI 5 · outside Madre de Dios

LATE · YEARLY

How much forest was lost

The yearly maps. 5 to 17 months late, but the reference for donors.

KPI 4 · forest lost

FIVE TESTS FOR EVERY KPI

Free and public | Arrives in time | Says where | Has a clear action | Honest about what it cannot see

HOW WE SET THE THRESHOLDS

One simple red line for each KPI

RULE 1

Clearly above normal

Gold price 20% above its average of the last 3 years.

KPI 1 · gold price

RULE 2

Average plus 2 standard deviations

Far above its own past. Normal ups and downs stay below the line.

KPI 2 · gold gap
KPI 4 · forest lost

RULE 3

More than the same quarters last year

More than 25% higher. In a reserve: one quarter worse than the worst whole year.

KPI 3 · FZS landscapes
KPI 5 · outside Madre de Dios

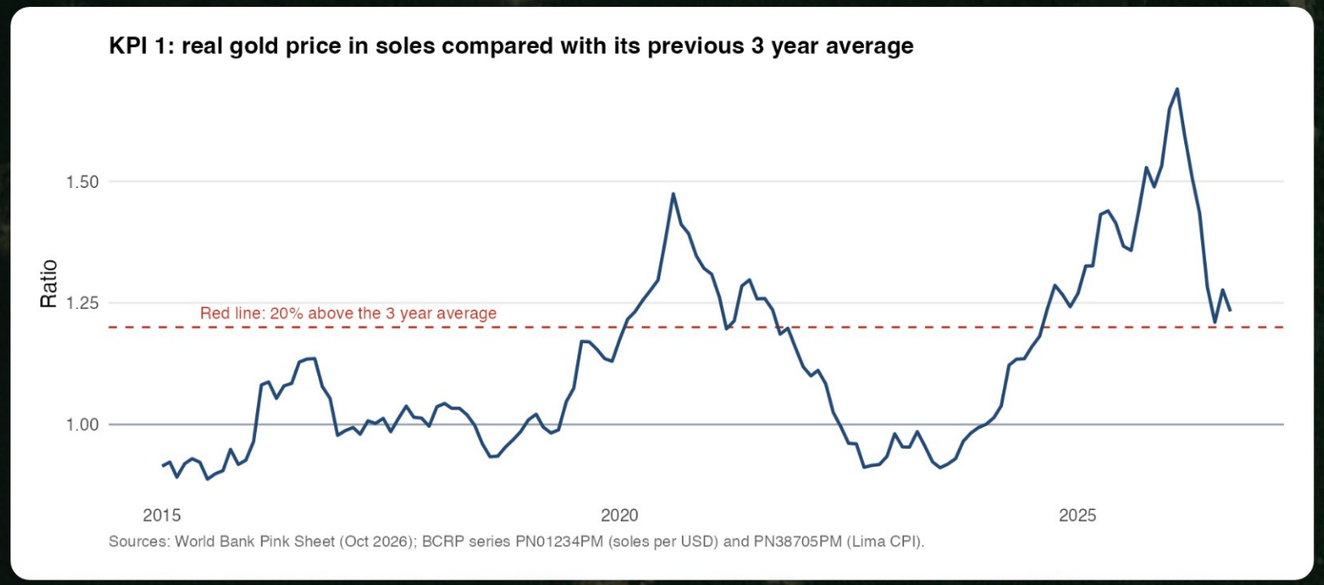
OUR ANSWER

Five KPIs for FZS

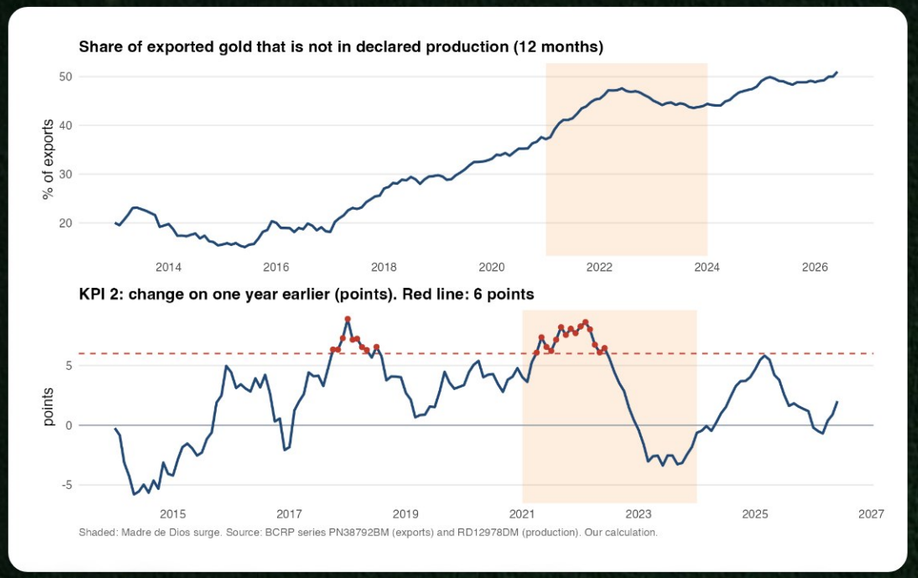
1	**Gold price in soles** compared with its average of the last 3 years	**Early** · monthly	RED 1.23 in Sep 2026
2	**Peru's gold gap** gold exported minus gold declared as produced	**Early** · monthly	NOT RED 51% of exports, stable
3	**New mining in FZS landscapes** hectares, Tambopata reserve, buffer zones, native communities	**Now** · quarterly	RED 966 ha inside the reserve
4	**Forest that became mining** reserves and buffer zones; all of Madre de Dios	**Late** · yearly	RED 175 ha in reserves, 2025
5	**New mining outside Madre de Dios** is mining spreading to new places?	**Now** · quarterly	NOT RED 11% lower this year so far

KPI 1 AND KPI 2 · EARLY

Gold pressure rose before the surge



Red today. Gold in soles is 23% above its 3 year normal. First red in February 2020, before the surge.



Half of Peru's exported gold is not declared as produced. The share jumped in April 2021. **Today it is stable: not red.**

> **Possible use:** planning patrols and flights for the next dry season.

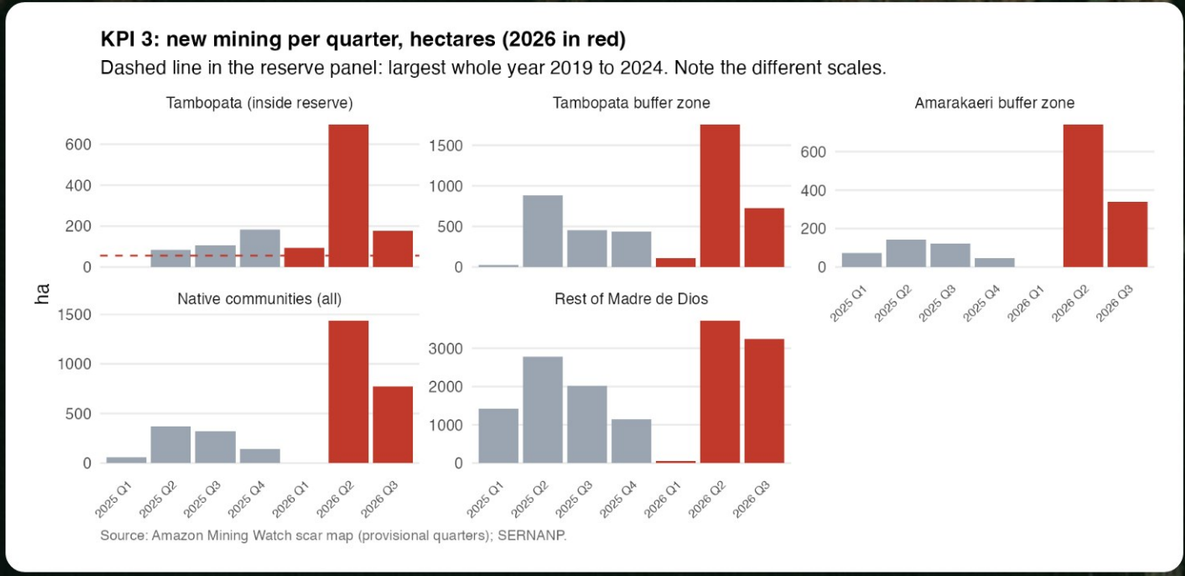
World Bank Pink Sheet; Central Bank of Peru (BCRP). Shaded: the 2021 to 2023 surge.

KPI 3 · NOW · TAMBOPATA NATIONAL RESERVE

Mining is moving into the Tambopata reserve

194 → 966 ha

New mining inside the reserve, first three quarters of 2025 and of 2026



Tambopata buffer zone	+89%
Amarakaeri buffer zone	+219%
Native communities	+196%
Rest of Madre de Dios	+12%

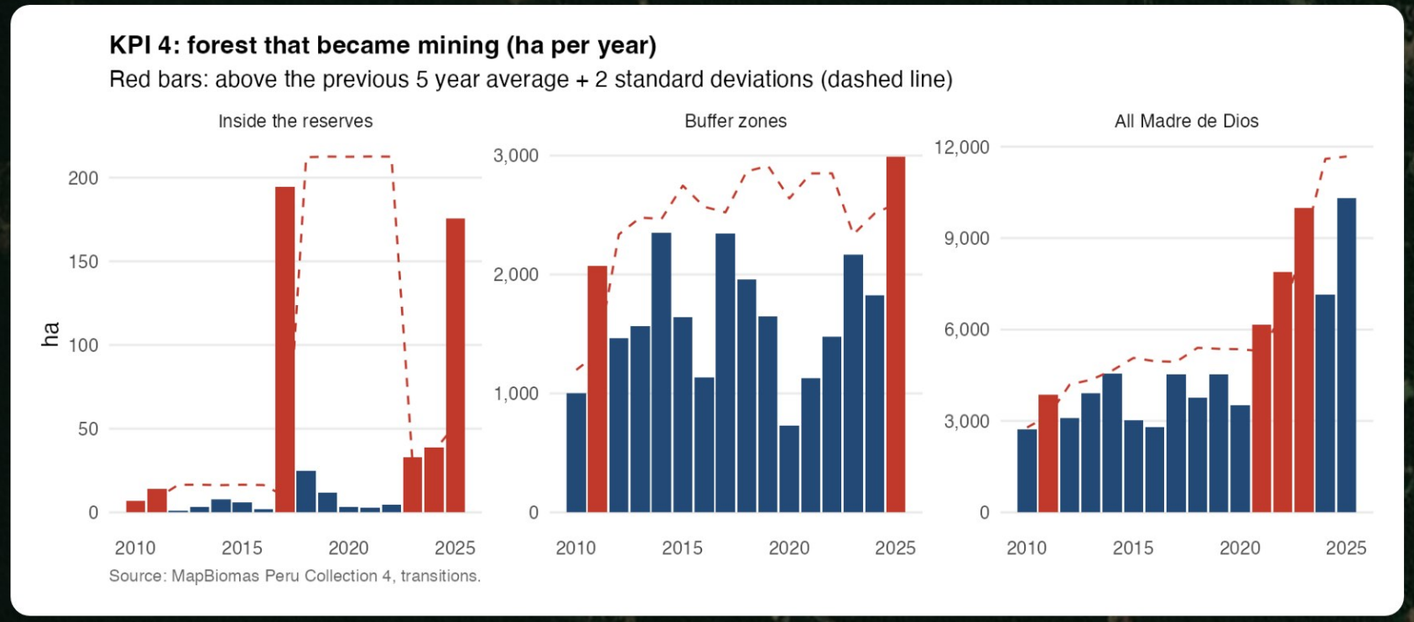
2025 and 2026 match MAAP and Global Conservation. 2026 Q3 is provisional.

Possible next step: share the locations with SERNANP and the environmental prosecutor.

Amazon Mining Watch scar map (3 Oct 2026, provisional quarters); SERNANP; MAAP #241; Mongabay Latam (27 Sep 2026)

KPI 4 · LATE · MAPBIOMAS YEARLY MAP

The yearly map confirms it inside FZS areas



Inside the reserves — 175 ha
red line 52 ha

Buffer zones — 2,992 ha
red line 2,595 ha

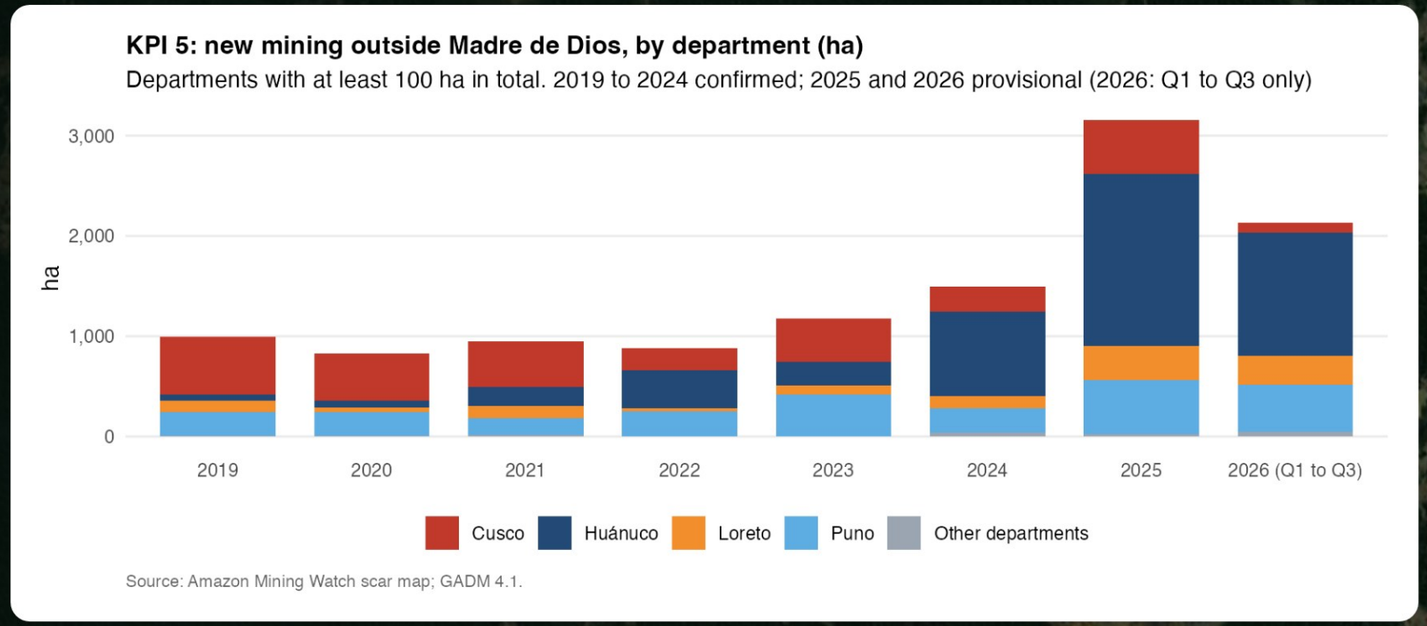
All Madre de Dios — 10,308 ha
record year, but the line moved up after the surge

> **For donor reports:** say "record year" in words, not only the colour.

MapBiomas Peru Collection 4, transitions. Reserves and buffer zones of Tambopata, Amarakaeri, Bahuaja Sonene and Manu.

KPI 5 · NOW · AMAZON MINING WATCH

Mining is spreading outside Madre de Dios



992 → 3,304 ha

2019 and 2025 (provisional)

Mostly in **Huánuco**, near El Sira.

65% is inside mining concessions, so we say "spreading", not "illegal".

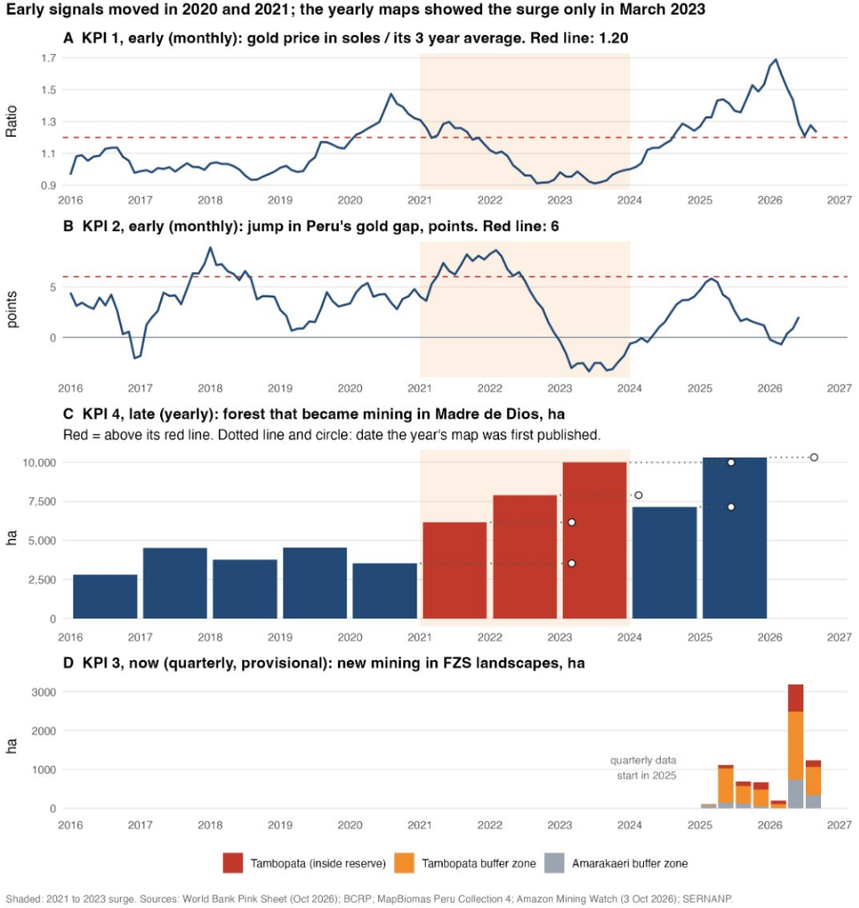
Today: not red (11% lower this year so far).

Worth watching: Huánuco, every quarter.

Amazon Mining Watch scar map; GADM 4.1; INGEMMET concessions

WOULD IT HAVE WARNED IN TIME?

Early signals came two years before the maps



KPI 1 · gold price red	Feb 2020
KPI 2 · gold gap jumps (public two months later)	Apr 2021
KPI 4 · first yearly map shows the surge	Mar 2023
KPI 3 · Tambopata reserve red	2025 Q2

> **An early signal shows pressure.** It is not a forecast of hectares.

> **It can be a false alarm.** The gold gap was also red in 2017 and 2018, with no surge after.

Same sources as KPI 1 to KPI 4: World Bank; BCRP; MapBiomas Peru; Amazon Mining Watch

CONTEXT, NOT KPIS

Other traces of illegal gold

BRAZIL · HEALTH

12,989

malaria cases caught in garimpos, 2025

Strong signal, but for Brazil only

COLOMBIA · RIVERS

117

dredges seized in the Amazon departments, 2025

Also measures police effort

PERU · MERCURY

176 t → 0

legal mercury imports, 2011 to today

Mercury is now smuggled, so trade data cannot see it

MADRE DE DIOS · COCA

42 → 1,492 ha

coca grown, 2019 to 2025

Same land as mining, but no tested link

MADRE DE DIOS · TOLLS

+68%

traffic at the mining toll, 2022 to 2025; other tolls almost flat

Only 3 tolls, and one stopped counting for 16 months

PERU · MACHINES

× 2.3

excavator imports, 2019 to 2024

Also used for roads and farms

Ministério da Saúde; MinDefensa; UN Comtrade; Devida and UNODC; OSITRAN (Likitha's analysis)

IDEAS WE DID NOT HAVE TIME TO TEST

What FZS could add next

A **Count turtle nests on river beaches**

Nesting beaches lost near mining. Our animal could become an indicator.

B **Use radar satellites**

They see through clouds, so the rainy months are not blind. SERNANP already runs one (RAMI).

C **Use FZS's own field reports**

Ranger patrols and dredge counts on Madre de Dios rivers.

D **Watch El Niño and drought**

El Niño years bring drought. Low rivers open more beaches and banks, so mining is easier. Erion's first test is not conclusive yet.

TO TAKE AWAY

Early signals, quarterly locations, yearly confirmation

Early numbers moved first

The gold price in 2020 and the gold gap in 2021. The maps came in 2023.

The problem is inside FZS areas now

Tambopata reserve: 194 to 966 ha in one year. The clearest case for action now.

Yearly maps for donors

Late but trusted. Say "record year" in words when the line moves up.

EVERY MONTH
KPI 1 and KPI 2

EVERY QUARTER
KPI 3 and KPI 5

EVERY YEAR
KPI 4, in the donor report

Thank you. Questions?